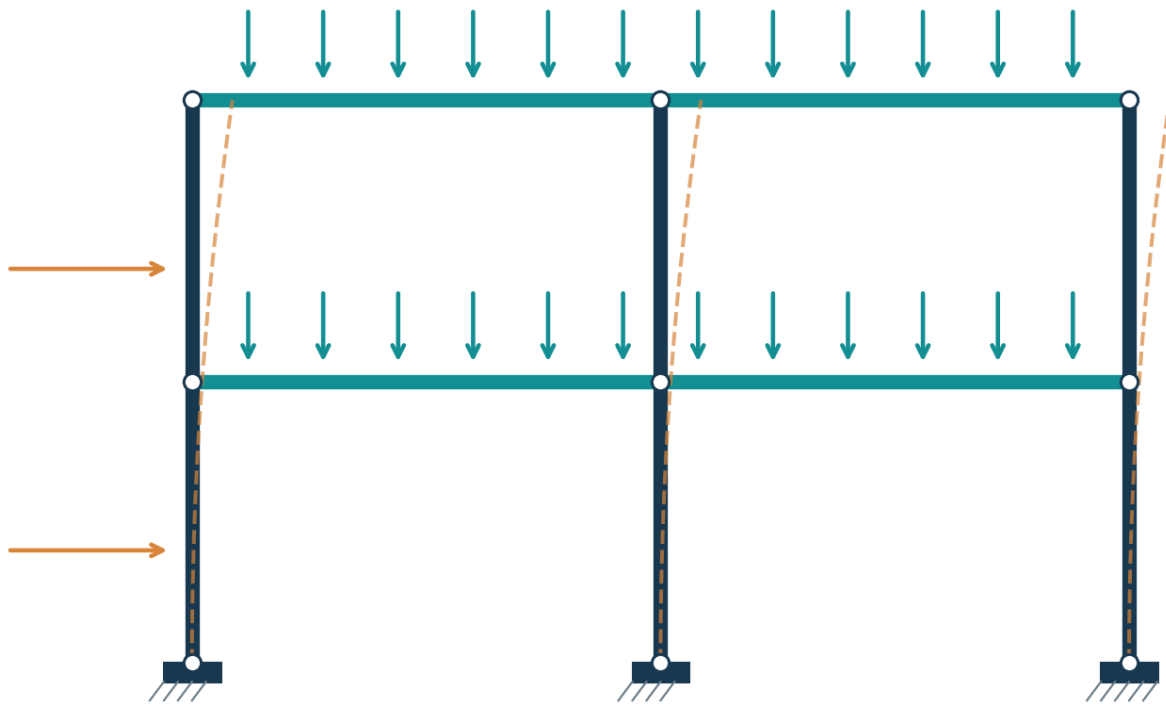


Reinforced Concrete Frame Design and Verification



*User manual and technical reference
Python and OpenSeesPy*

beta_03 • English edition • 5 October 2026



Planar elastic frame with gravity and lateral actions

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This volume covers modelling, analysis and verification of reinforced concrete frames, including manual revision of sections and reinforcement. Automated optimisation is reserved for a subsequent volume.

About this manual

This manual explains how to prepare and verify a planar reinforced concrete frame using the beta_03 code supplied with Polito Work Station. It follows the active modules in src/framekit and the console sequence in main.py. The intended reader is an engineer or student familiar with frame analysis and reinforced concrete design.

Design and verification in this volume

For verification, enter the geometry, reinforcement, materials and actions of the frame, then inspect the calculated demands and utilisation indices. For design, choose an initial section and reinforcement arrangement, run the same checks and revise the inputs manually. Section presets provide starting values; they are not automatically compliant designs.

The model is elastic and planar. The implemented EC2 member checks and EC8 drift checks cover specific procedures described here. A positive result is limited to those procedures and the load cases actually completed. Detailed reinforcement design and the additional checks identified on page 15 remain the designer's responsibility.

Reading guide

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Edition and evidence

The beta_03 regression suite passed 33 tests with Python 3.12.14 and OpenSeesPy 3.8.0.0 on Linux. It includes analytical beam benchmarks, equilibrium, section signs, masses, modal combination and dynamic drift tests. The test log is docs/test_results_beta_03.txt. These checks support software verification, rather than a complete validation for every structural application.

1 Frame analysis workflow

Complete the static workflow first. Add dynamic analyses only after geometry, action signs, section data and equilibrium have been checked. The return arrows below describe manual engineering revisions, each followed by a new analysis session.

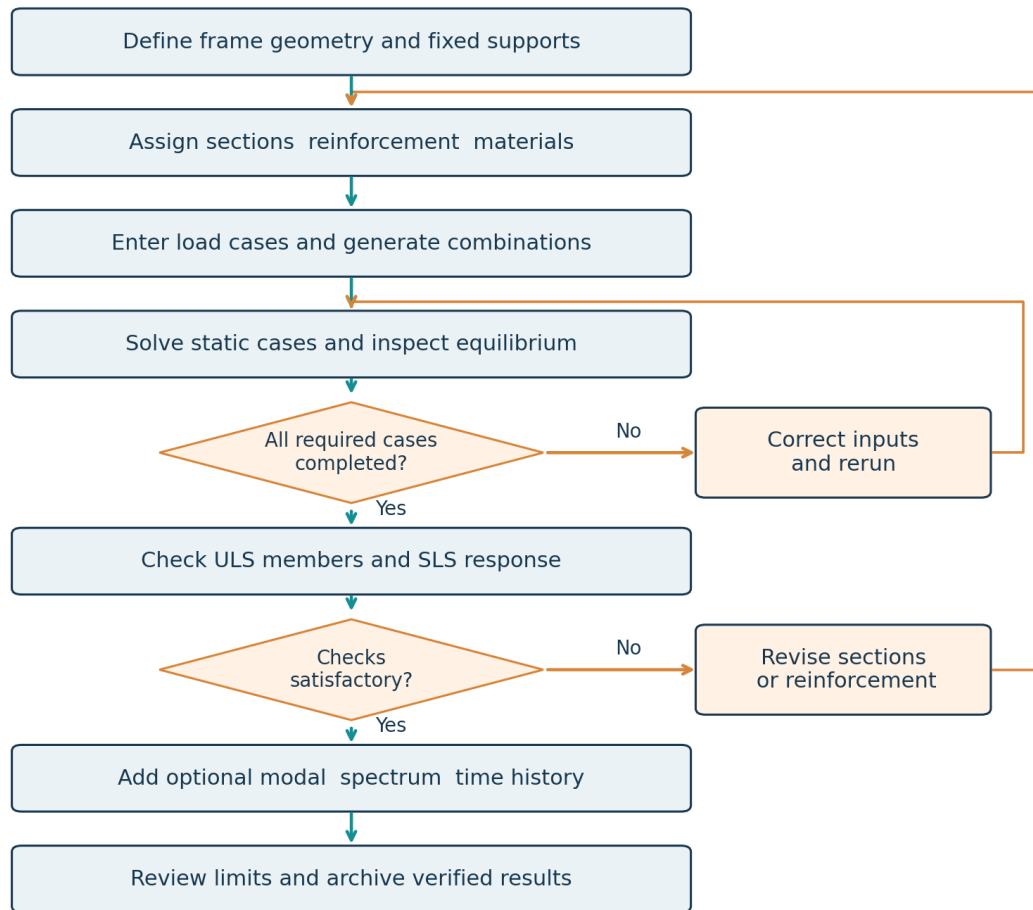


Figure 1 Workflow for RC frame design and verification. Revision loops are manual.

Console choices for this volume: select rc, answer n when asked to add shear walls, and answer n to “Run retrofit optimization?”. The optional global performance summary can consolidate the analyses already performed. It does not create missing assessments.

2 Installation and execution

Extract the beta_03 archive and open a terminal in the directory containing main.py. Keep main.py and src together. Create an isolated Python 3.12 environment before installing dependencies.

Windows command prompt

```
py -3.12 -m venv .venv
.venv\Scripts\activate
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
```

Linux or macOS terminal

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
```

requirements.txt defines the release dependencies. requirements-tested.txt records the exact Linux versions used for verification, including the platform-specific OpenSeesPy wheel. Use the former for ordinary installation and the latter to compare environments.

Confirm the installation and run

```
python -c "import openseespy.opensees as ops; print(ops.version())"
python main.py --version
python main.py
```

A successful import confirms that the solver library loads. It does not verify a structural model. The console displays questions and opens Matplotlib figures; close each figure to continue if the next question does not appear.

Organise a case

```
python main.py --output-dir results/frame_case_01
python -m pip freeze > requirements_case_01.txt
```

By default each run creates results/run_TIMESTAMP. Reusing an explicit output directory can overwrite files with the same names. Store the input choices, console log, units, dependency versions and output tables together. Earlier results in archive/original_results are historical records, not results of the current session.

Directory	Use
src/framekit	Active analysis and verification modules
docs	Manual, figures, change record and test logs
tests	Regression tests
data/records	Acceleration files with separate metadata
results	Outputs from new analysis sessions

3 Geometry and numbering

Choose rc at “Structure type”. Enter the number of floors, a common floor height, the number of bays and the length of each bay in metres. Different bay lengths are allowed. The ordinary dialogue uses a uniform storey height and automatically fixes all three degrees of freedom at the base nodes.

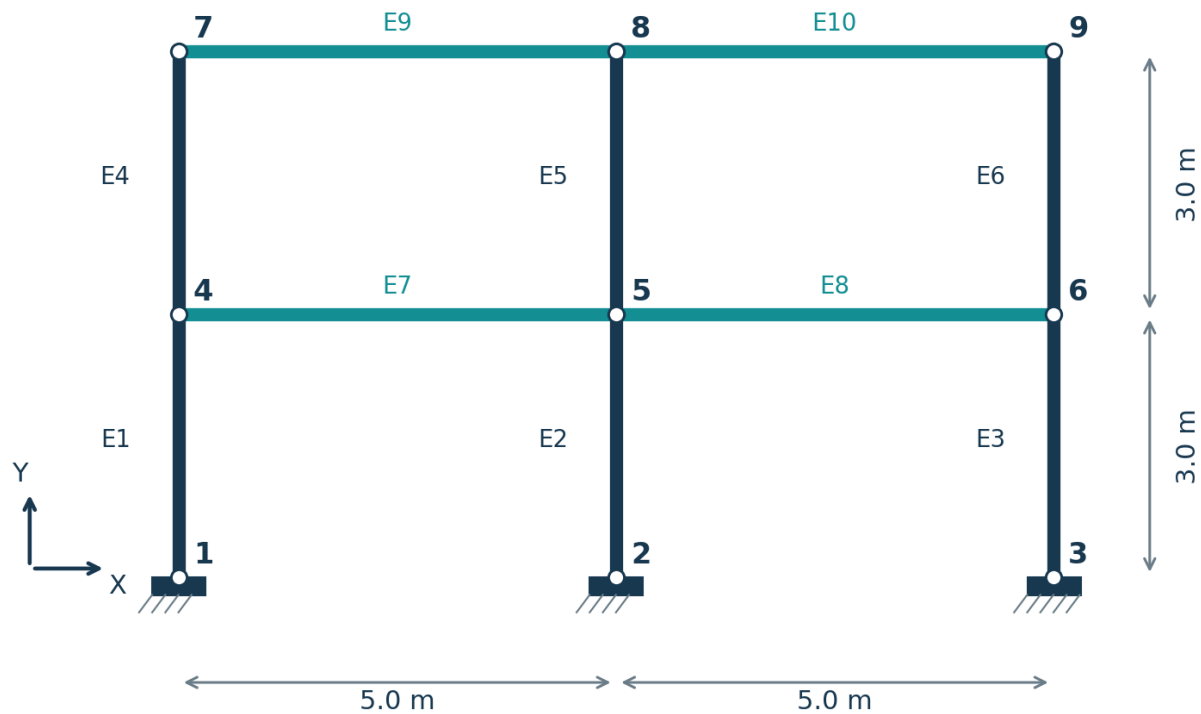


Figure 2 Two storeys and two bays. Circles mark nodes, not hinges; E labels identify elements.

Node and member identifiers

Nodes are numbered from left to right at the base, then at successive floors. For n_b bays, level f and grid position j , the node ID is $f(n_b + 1) + j + 1$, with f and j starting from zero. Columns are numbered first, storey by storey; beams follow from the first elevated floor upwards.

In Figure 2 there are nine nodes and ten members. Nodes 1–3 are fixed. Elements 1–6 are columns; 7–8 are first-floor beams and 9–10 are roof beams. Use these element IDs when assigning extra distributed loads. Point loads use node IDs.

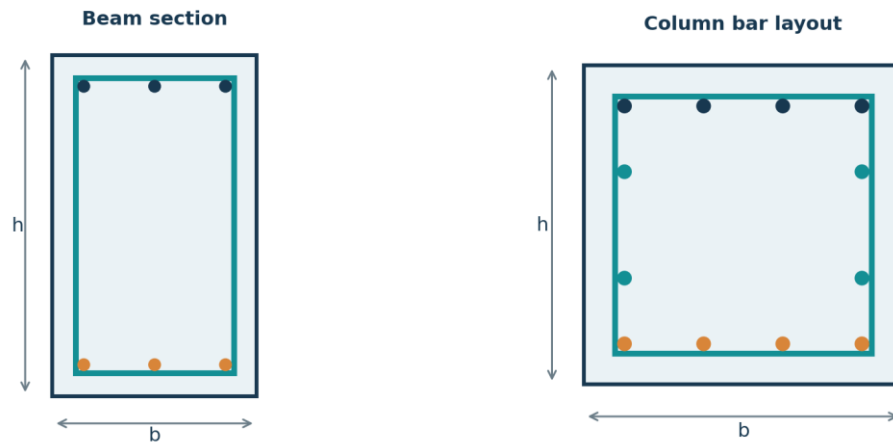
Modelling decisions

Each node has horizontal displacement, vertical displacement and in-plane rotation. The global axes are X to the right and Y upwards. A planar frame cannot represent out-of-plane bending or building torsion. Alternative base springs, support releases and variable storey heights require an explicit source-level modelling change.

The horizontal rigid diaphragm option is selected before static analysis and retained for the other workflows. Answer n for the introductory example. Answer y only when equal horizontal floor displacements are an appropriate assumption; this option ties horizontal translations, rather than modelling a slab or a three-dimensional diaphragm.

4 Sections and reinforcement

RC SECTION INPUT provides library selection, default presets and manual entry. Option 2 selects beam B3 and column C3. Initial beam and column sections are common to their respective member groups. For the frame workflow in this volume, revise these selections manually and rerun the analysis as necessary.



Top and bottom layers include corner bars. Side bars exclude corners. Column layout is schematic.

Figure 3 Rectangular sections and bar conventions. The column side bars illustrate optional input.

Field	Input and meaning
b_mm h_mm	Width and depth in mm
cover_mm	Cover to the outside of the stirrup, in mm
n_top n_bot	Bar counts on top and bottom faces, including corners
phi_top_mm phi_bot_mm	Top and bottom longitudinal bar diameters, in mm
n_left n_right	Intermediate side bars, excluding corner bars
phi_side_mm	Diameter of intermediate column side bars, in mm
stirrup_phi_mm	Stirrup or tie diameter, in mm
stirrup_spacing_mm	Ordinary longitudinal spacing, in mm
stirrup_legs	Number of resisting stirrup legs
stirrup_spacing_critical_mm	Declared spacing in end zones, in mm

Cover and bar positions

The distance from the concrete surface to a longitudinal bar centre is cover + stirrup diameter + half the longitudinal bar diameter. `section_geometry.py` supplies shared bar coordinates for stiffness and resistance calculations. Inspect the section drawing to detect repeated corner bars, misplaced side bars or an unintended top–bottom arrangement.

The critical-zone spacing is reported but does not automatically generate a spatially varying shear resistance. Bar clearance, anchorage, laps, joints and the complete detailing rules require separate design checks.

5 Materials and elastic properties

Two distinct input blocks

The material dialogue asks for concrete Young's modulus E in MPa and density in kg/m^3 . The EC2 dialogue separately supplies concrete and reinforcement strengths. Changing E does not update f_{ck} . The steel modulus used by main.py for transformed elastic properties is 200 GPa.

Parameter	Default or example	Purpose
Concrete E	30000 MPa in the example	Elastic stiffness
Concrete density	2500 kg/m^3 in the example	Self weight and derived masses
Reinforcement E_s	200000 MPa	Transformed section stiffness
Concrete class	C25/30 by default	Characteristic $f_{ck} = 25$ MPa
Reinforcement	B500 by default	Characteristic $f_{yk} = 500$ MPa
γ_c γ_s α_{cc}	1.50 1.15 0.85	Design strength coefficients

The default design strengths are $f_{cd} = \alpha_{cc} f_{ck} / \gamma_c = 14.17$ MPa and $f_{yd} = f_{yk} / \gamma_s = 434.78$ MPa. Custom material strengths and coefficients can be entered in the EC2 dialogue. Their suitability depends on the chosen design basis and applicable national provisions.

Transformed uncracked section

The function `section_properties_from_fiber` constructs equivalent elastic properties from the concrete rectangle and all longitudinal bars, including intermediate side bars. Despite its name, this workflow creates `elasticBeamColumn` elements; it does not define nonlinear fibre beam elements.

$$n = E_s / E_c$$

$$A_{eq} = A_c + (n - 1) \sum A_{s,i}$$

$$y_c = [A_c h/2 + (n - 1) \sum (A_{s,i} y_i)] / A_{eq}$$

$$I_{eq} = I_g + A_c(h/2 - y_c)^2 + (n - 1) \sum [A_{s,i}(y_i - y_c)^2]$$

Here $A_c = bh$ is the gross concrete rectangle, $I_g = bh^3/12$, $A_{s,i}$ is one bar area and y_i its coordinate through the depth. The factor $n - 1$ replaces concrete displaced by reinforcement. Convert section dimensions to metres when calculating model properties; the solver uses E in Pa, A in m^2 and I in m^4 .

Consequences for interpretation

The analysis uses uncracked transformed stiffness, a Linear geometric transformation and elastic members. It does not predict cracking progression, yielding, plastic redistribution, creep, shrinkage or P Delta effects. Where effective cracked stiffness or second-order response governs design, an appropriate additional model is required. Selecting Newton as the solution algorithm does not change these constitutive assumptions.

6 Actions and combinations

The solver uses metres, newtons, kilograms and seconds. The input interface converts section mm to m, E in MPa to Pa, forces in kN to N and distributed actions in kN/m to N/m. Use negative global vertical values for downward actions.

Case	Definition and entry
SW	Automatic beam and column self weight $\rho g A$ with $g = 9.81 \text{ m/s}^2$
G	Additional permanent floor and roof actions; extra G loads allowed
Q	Variable actions entered through Extra loads
W+	Rightward action on external left columns
W-	Leftward action on external right columns

Extra loads accept uniform, point or triangular. A point load acts at a node. Distributed loads use element IDs, including lists such as 2,3,10-12 or all. "Triangular" accepts end intensities at i and j and also represents a trapezoid. Confirm the assigned member direction and inspect the load plot before solving.

Wind intensity varies with height, but each column storey receives a uniform value evaluated at its mid-height. Thus the implemented storey load is a piecewise uniform representation, rather than an exactly triangular load on each column.

Nine implemented static combinations

Family	Expression	Cases
ULS Q leading	$1.35 \text{ SW} + 1.35 \text{ G} + 1.50 \text{ Q} + 0.90 \text{ W}\pm$	2
ULS W leading	$1.35 \text{ SW} + 1.35 \text{ G} + 1.05 \text{ Q} + 1.50 \text{ W}\pm$	2
SLS characteristic	$\text{SW} + \text{G} + \text{Q} + 0.60 \text{ W}\pm$	2
SLS frequent	$\text{SW} + \text{G} + 0.50 \text{ Q}$	2
SLS quasi permanent	$\text{SW} + \text{G} + 0.30 \text{ Q}$	1

The two frequent combinations coincide because the implemented wind coefficient $\psi W2$ is zero. Factors are fixed in constants.py. Four combinations are ULS and five are SLS, including the duplicate frequent cases.

Design basis to record

The generated set does not automatically cover all spatial live-load arrangements, favourable permanent actions, use-category-specific combination factors or seismic member combinations $G + \psi Q \pm E$. Define the required action set for the project and supplement the implementation where necessary. SW is already generated: avoid entering the same frame self weight again in G.

7 Static analysis and internal forces

The model is rebuilt for every combination. The static solver uses Transformation constraints, RCM numbering, ProfileSPD, Newton and a LoadControl step of 1.0. Distributed actions are transferred to OpenSees with beamUniform, including its variable-load form.

Establish completeness before reading envelopes

The console lists successful and failed combinations. Confirm nine completed static combinations and four completed ULS member assessments for the default set. main.py can display or export results from the cases that succeeded; the presence of a table alone does not prove that every required case was included. A failed solver return must be treated as a missing assessment.

Force reconstruction along each member

member_forces.py reconstructs N, V and M from end forces and applied distributed loads. Internal N is positive in tension, sagging M is positive and $V = dM/dx$. The EC2 output field NEd_signed_kN uses the opposite axial sign, positive in compression; MEd_signed_kNm retains the sagging-positive convention. Demand magnitude fields are nonnegative.

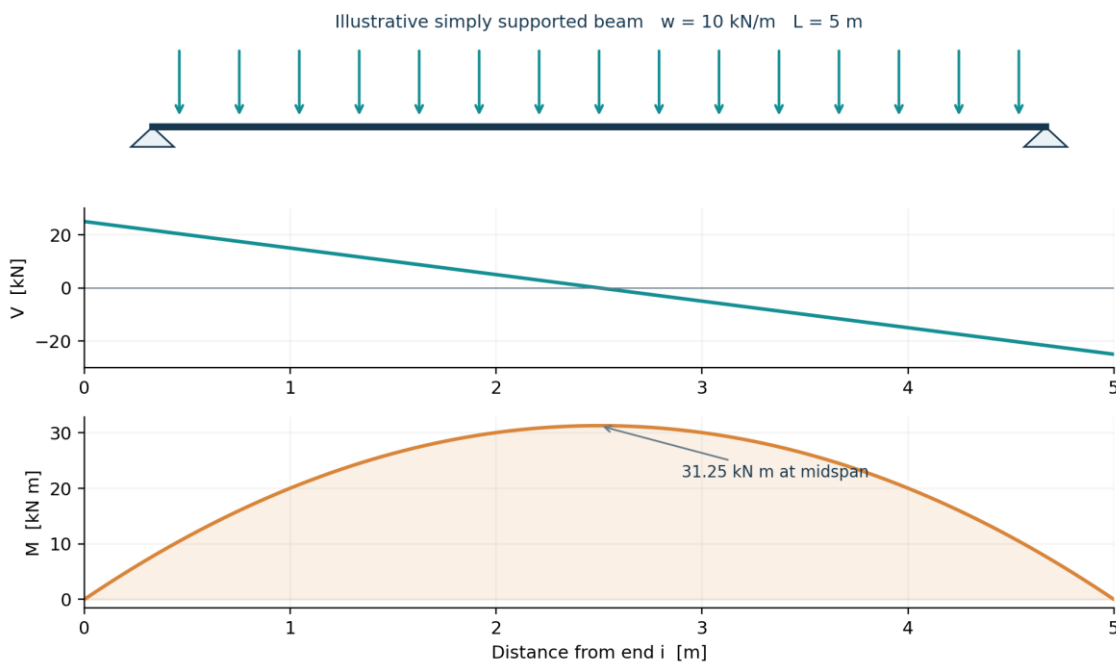


Figure 4 Analytical illustration of an internal maximum. This simply supported benchmark is not the frame example.

For the illustrated beam, support reactions are 25 kN and the maximum moment is $wL^2/8 = 31.25 \text{ kN m}$. Reading only member ends would miss this moment. The beta_03 checks sample at least 21 stations and add analytical extrema of the force functions, preserving simultaneous N, V and M at each station.

The N/V/M menu offers one combination, all combinations or no diagrams. First inspect a single governing ULS case. Check reactions against total applied actions and review the expected force pattern before interpreting a utilisation map.

8 Member resistance checks

Answer y to “Run EC2 checks for ULS combinations?”. The program solves the four ULS cases again and exports a worst-utilisation table for each member. Each governing row corresponds to one combination and one location along the member.

Beam checks

Beam bending resistance uses the tensile reinforcement on the side selected by the moment sign. The steel tensile force is limited consistently with the maximum concrete compression block. Axial tension resistance is based on steel; axial compression is also checked. The beam utilisation is the maximum of η_V , η_M and η_N . This component-based procedure is not a complete beam N M interaction analysis.

Column checks

The column routine samples a strain-compatible N M resistance domain using all longitudinal bars and both bending signs. Pure bending resistance is interpolated at $N = 0$, while the axial tension and compression limits are included explicitly. η_{NM} represents the implemented interaction measure; the final column η is $\max(\eta_V, \eta_{NM})$. Domain discretisation remains an approximation.

Shear resistance

The beta procedure evaluates simplified concrete and stirrup contributions and applies an explicit compression-strut limit at $\theta = 45^\circ$. The ordinary stirrup spacing is used in the resistance model. The declared end-zone spacing and the complete shear detailing requirements need separate review. Compare the implemented resistance assumptions with the adopted code provisions before project use.

Field	How to read it
η_V	Shear demand divided by implemented shear resistance
η_M	Bending utilisation; auxiliary for a column
η_N	Axial utilisation; also included in the beam governing index
η_{NM}	Beam bending index or column interaction index
η	Governing combined utilisation for the implemented checks
Governing_x_m	Distance from member end i at the governing station
Combination	Load combination producing the governing row
StaticPass	OK for $\eta \leq 1$; FAIL for $\eta > 1$

Limits of the member result

An OK flag applies to the resistance model and evaluated cases. It does not include beam–column joint resistance, anchorage, lap design, ductility detailing, capacity design, column instability or complete SLS checks. A governing row retains simultaneous demands; its MEd value need not equal the independent maximum moment over the member.

9 Outputs and serviceability review

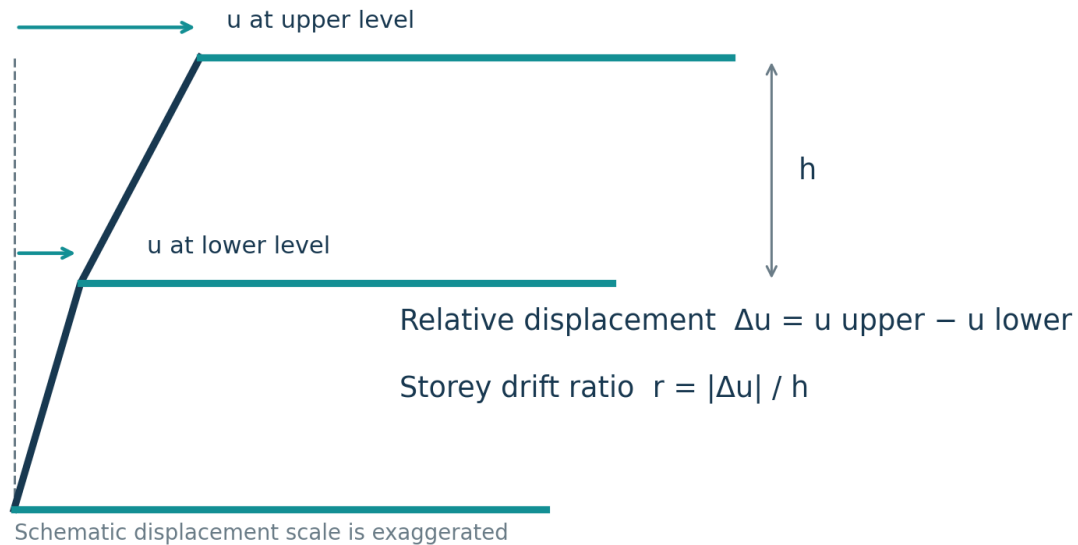


Figure 5 Storey drift uses relative horizontal displacement between adjacent levels.

At every vertical grid line, the static routine calculates $|u_{\text{upper}} - u_{\text{lower}}| / h$ and retains the maximum for the storey. Roof displacement and storey drift describe different response quantities. A deformed-shape plot uses an amplified scale; read the displacement values from the table.

Output	Information and review
story_drifts.csv	Combination, storey, drift in m and mm, ratio, percent and controlling nodes
ec2_worst_utilization.xlsx	Worst ULS utilisation for each element, governing location and demand/resistance fields
N/V/M figures	Member force distribution for the chosen combination
performance_summary.xlsx	Optional consolidation of performed static and dynamic assessments

Using the coloured results

EC2 utilisation colours change at 0.70, 0.85 and 1.00; the numerical eta field determines the resistance outcome. Static drift graphics also contain colour bands, but they do not select a project-specific serviceability criterion. Identify the combination, deformation measure and allowable limit explicitly.

SLS combinations produce elastic responses. Cracked-section deflection, long-term effects, crack width and reinforcement stress limits are not automatically checked by generating these combinations. For a serviceability decision, calculate the additional quantities and stiffness assumptions required by the adopted design basis.

What to archive

Record the input sections and reinforcement, support and diaphragm assumptions, action set, successful combinations, output units and unassessed checks. Keep the whole result directory and console log. A missing dynamic result should remain NOT_ASSESSED; it cannot be inferred from a static OK flag.

10 Optional modal and spectrum analysis

Answer y to “Go to dynamic analysis now?” after completing the static review. These routines assess elastic in-plane response in the horizontal direction of the current frame. DOF 1 is horizontal; DOF 2 is vertical and is rejected for excitation in this workflow. Identical X/Y frame clones and their directional combination are disabled.

Mass definition and modes

Masses are derived from $SW + G + \psi Q$, with $\psi = 0.30$ initially, plus any explicitly entered lumped floor mass. The beta includes tributary column weights as well as beam weights. Weight assigned to restrained base nodes does not participate in vibration. Upward vertical actions cannot be treated as positive seismic weight and are rejected by the mass routine.

The principal mass is assigned to horizontal translation; small numerical masses are used for the other DOFs. Keep the diaphragm choice made before static analysis. Select a feasible number of modes and inspect periods, shapes and cumulative participating mass. Numerical modes associated with small masses require scrutiny. The outputs are modal_periods.csv and modal_participation.csv.

Elastic response spectrum

Input	Entry
ag	Peak ground acceleration in g
Ground class	A to E
Spectrum type	1 or 2
Damping ratio	Fraction; enter 0.05 for 5 percent
Importance factor	Project value entered separately
Direction	1 for the current horizontal frame
Modal combination	CQC or SRSS

The routine constructs a piecewise elastic spectrum and keeps modal indices aligned with their periods and spectral ordinates. The dialogue does not provide a behaviour-factor q reduction for a design spectrum. CQC includes modal correlation; SRSS is the square root of the sum of squared modal responses.

For drift, the routine first forms relative floor displacement within each mode and then combines the modal contributions. Subtracting independently combined nodal displacement magnitudes would lose this relationship. Outputs include response_spectrum_results.xlsx and response_spectrum_drifts.csv.

Implemented damage limitation drift check

The routine uses $dDLS = v dRSA$, initially $v = 0.50$, and compares $dDLS/h$ with 0.0050 for brittle attached components, 0.0075 for ductile attached components or 0.0100 for detached/non-interfering components. It exports ec8_drift_check_single_direction.xlsx. Confirm v and the limit against the project requirements. This drift check does not automatically perform seismic member resistance checks under $G + \psi Q \pm E$.

11 Optional linear time history

The time-history menu accepts one or more acceleration records. Use direction 1. Each record needs a path, sampling interval dt_record in seconds, scale factor and acceleration unit g or m/s^2 . Scaling and unit conversion are distinct operations.

Prepare and identify the input

```
# Single acceleration column in m/s2
0.000
0.120
-0.080
0.050
```

The example above shows file format only, rather than a seismic record. A time–acceleration file must be converted to one acceleration column. The reader accepts `#` comments but rejects multiple columns, NaN and infinite values. Keep station, component, event, source, sampling interval and units in the case record.

`data/records/elcentro.txt` preserves the original numeric samples, but its sampling interval, units and provenance are not verified. Do not infer these values from the filename. The release demonstration contains a synthetic record with explicit metadata; its purpose is a software exercise.

Damping and time integration

The routine calibrates Rayleigh damping from two selected modal frequencies. The initial mode choices are $i = 1$ and $j = 3$; change them if fewer modes are available. Use two distinct valid frequencies and enter damping as a fraction.

```
 $\alpha = 2 \xi \omega_i \omega_j / (\omega_i + \omega_j)$ 
 $\beta = 2 \xi / (\omega_i + \omega_j)$ 
ops.rayleigh( $\alpha$ , 0, 0,  $\beta$ )
```

The analysis step can differ from the sampling interval. Record duration is $(\text{number of samples} - 1) dt_record$, with a shortened final step where necessary. Each record restarts from the initial state and time zero. Review the simulated duration, convergence status, roof displacement, dynamic base shear and peak storey drifts.

Batch outputs

File	Content
time_history_batch_summary.csv	Record metadata, completion and global response metrics
time_history_batch_peak_drifts.csv	Peak storey drift for each record
time_history_batch_envelope_drifts.csv	Storey maxima across successful records
time_history_batch_results.xlsx	Combined batch tables

Failed records remain in the diagnostic summary and are excluded from the successful drift envelope. Maxima from different records or storeys are not simultaneous response states. These analyses remain elastic and do not estimate plastic damage or collapse capacity.

12 Worked input example

Use this small example to learn the dialogue, load mapping and deterministic checks. It is a trial frame, rather than a proposed compliant design. Figure 2 gives its numbering.

Input or prompt	Choice
Structure type	rc
Floors and floor height	2 floors; 3.0 m
Bays and lengths	2 bays; 5.0 m and 5.0 m
Add shear walls	n
Section input option	2; retain B3 beam and C3 column
EC2 material option	1; retain default coefficients
Concrete E and density	30000 MPa; 2500 kg/m ³
G first floor and roof	-8 kN/m; -6 kN/m
Top wind W+ and W-	2 kN/m for each direction
Extra Q uniform vertical	Elements 7-8: -5 kN/m
Next extra Q uniform vertical	Elements 9-10: -3 kN/m
Horizontal rigid diaphragm	n
Run EC2 checks	y
N/V/M menu	1; choose a ULS combination
Go to dynamic analysis	n for the introductory session
Build performance summary	y
Run retrofit optimization	n

Expected input consistency checks

The frame has 9 nodes, 6 columns and 4 beams. B3 is 300 × 500 mm and C3 is 400 × 400 mm. Self weight is 3.67875 kN/m for beams and 3.924 kN/m for columns, using $\rho = 2500 \text{ kg/m}^3$ and $g = 9.81 \text{ m/s}^2$. Unfactored frame self weight is 144.207 kN; G totals 140 kN and Q totals 80 kN.

For W+, the first- and second-storey intensities are +0.50 and +1.50 kN/m on the left external columns; W- applies negative values to the right external columns. Each directional case has a horizontal resultant of magnitude 6 kN. Check the action plots and reaction sums with the relevant combination factors.

The default set contains nine static combinations and four ULS checks. Periods, displacements and utilisation indices must be obtained from the run and reviewed within the model assumptions; they are not supplied here as certified design values. If a section is revised, regenerate its weight and rerun the complete assessment.

13 Manual design review and limits

Revise a frame through engineering decisions

Start from the load path, spans, storey heights and practical section sizes. Choose longitudinal and transverse reinforcement that can be detailed and constructed. After analysis, identify the governing combination, member location and resistance mechanism. Revise the relevant inputs, then repeat all affected static and dynamic analyses.

Increasing section depth changes stiffness, self weight, force distribution and potentially seismic mass and period. Adding reinforcement changes stiffness and resistance, while reduced stirrup spacing primarily affects shear resistance and detailing. Reanalyse the whole frame after a change; a local capacity increase alone does not establish the final frame response.

What the beta checks cover

Assessment	Implemented scope	Additional review
Global frame analysis	Planar elastic members and linear geometry	Cracking, second order, out-of-plane effects
Beam ULS	Bending sign, axial and shear components	Complete interaction and detailed reinforcement rules
Column ULS	Sampled N M domain and shear	Slenderness, second order and ductility
SLS response	Elastic displacement and storey drift	Crack widths and short/long-term deflection
Dynamic response	Modal, elastic spectrum and linear TH	Seismic member combinations and capacity design
Frame details	Input bar arrangement and spacing	Joints, anchorage, laps, foundations and connections

Before issuing an assessment

Confirm action categories and combination factors, all required solver completions, reaction equilibrium and realistic deformation patterns. Confirm that reinforcement positions and moment signs agree with the resistance checks. Review every storey and member, rather than only the most favourable graphic.

Document the completed checks and the remaining ones explicitly. An EC2 StaticPass of OK cannot stand in for an EC8 drift assessment, seismic resistance, serviceability verification or a foundation design. NOT_ASSESSED means that the required evidence is missing. The 33 software regression tests do not eliminate the need for independent verification of the project model.

Scope of this edition

This volume is limited to RC frame modelling, analysis, verification and manual design revision. Automated intervention selection, jacking searches, cost models and optimisation datasets will be documented separately. Their modules remain in the beta_03 distribution, but are not part of the workflow described here.

14 Code map and references

Module in src/framekit	Responsibility
data_models.py geometry.py	Input data structures, nodes and members
section_library.py section_geometry.py	RC presets, input, bar coordinates and drawings
loads.py combinations.py	Actions, self weight and static combinations
constants.py units.py	Factors and input/solver conversions
opensees_model.py	Elastic model assembly, constraints and solver
static_analysis.py	Nodal displacement and storey drift
member_forces.py diagrams.py	Internal force functions, stations and diagrams
ec2_checks.py	Member resistance, utilisation and exports
dynamics.py	Masses, eigenvalues and modal participation
response_spectrum.py	Elastic spectrum and modal combination
ec8_drift_checks.py	Implemented damage limitation drift checks
time_history.py	Record reading and linear transient response
performance_summary.py	Aggregation of completed assessments

Sources for this manual

The primary source is the supplied beta_03 code: main.py, src/framekit and the release tests. docs/CHANGELOG_beta_03.md records the numerical changes; docs/source_changes_beta_03.patch records source edits. Existing historical reports are kept separately in archive/original_results.

External technical documentation

The following official resources were checked on 5 October 2026. They explain the underlying solver command and Python environment; the release source remains authoritative for the program's particular implementation.

OpenSeesPy elasticBeamColumn element documentation

Python 3.12 Virtual Environments and Packages

ARTISTE Research Group at Politecnico di Torino

Release identification

Software version beta_03. English manual focused on RC frames. Authors: Arman Hajiha, Giuseppe Carlo MARANO, ARTISTE Research Group. The manuscript documents the supplied code and does not claim complete compliance with a particular Eurocode edition or National Annex. Record the adopted normative basis separately for each project.